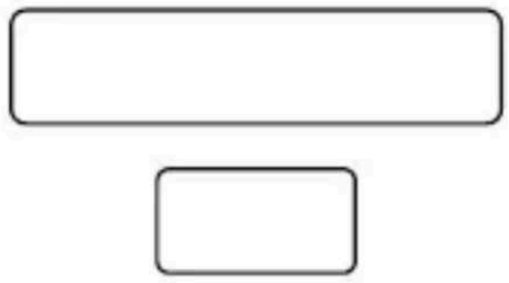
Aakash

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$$\frac{V^2}{R}$$

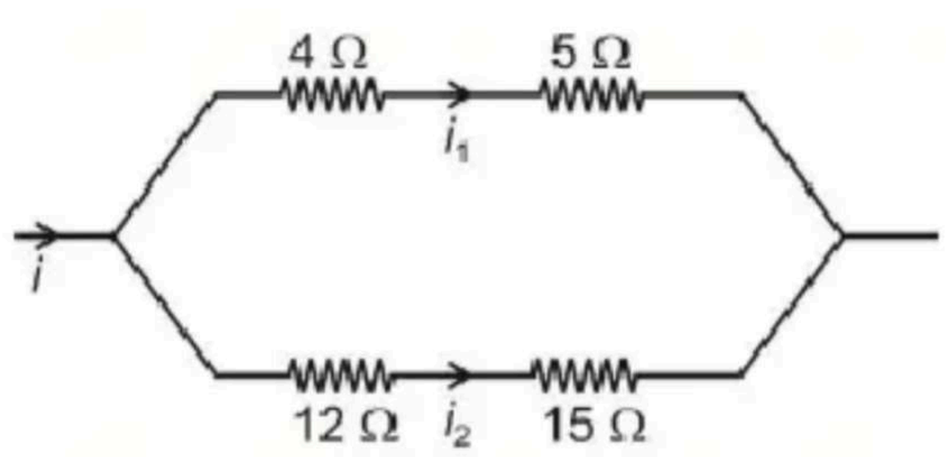
$$P \propto \frac{1}{R}$$

$$\frac{R_Q}{R_P} = \frac{P_P}{P_Q} = \frac{400}{100} = \frac{4}{1}$$

$$P_1 = \frac{V^2}{2R}$$

$$R_{\text{Net}} = \frac{R}{2}$$

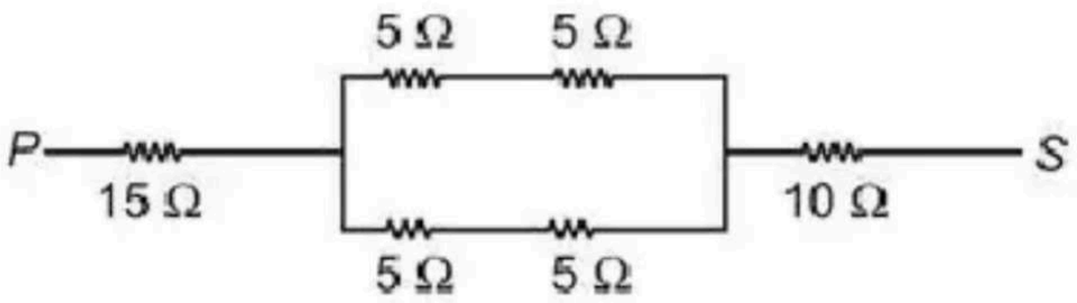
$$P_2 = \frac{V^2}{\left(\frac{R}{2}\right)} = \frac{2V^2}{R} = 4P_1 = 400 \text{ W}$$



$$i_1 = i \times \frac{(27)}{(27+9)}$$

$$i_2 = i \times \frac{(9)}{(27+9)}$$

$$i_1 : i_2 = 27 : 9 = 3 : 1$$



$$i = \frac{10}{5} = 2 \text{ A}$$

$$V_{\text{A}} + 4 \times 1 - 2 \times 1 = V_{\text{B}}$$

$$(V_A - V_B) = -2V$$

$$K = I\rho = I \times \frac{R}{L} = \frac{5 \times 4}{1} = 20$$

$P = \frac{V^2}{R}$

$\frac{\Delta P}{P} = \frac{2\Delta V}{V}$

$\frac{\Delta P}{P}\% = 2 \times 3 = 6\%$

(9) **Answer :** (4)
Solution:
For parallel.

$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2}$

$\frac{1}{\frac{R_0}{2}(1+\alpha_p T)} = \frac{1}{R_0(1+\alpha_1 T)} + \frac{1}{R_0(1+\alpha_2 T)}$

Using binomial expansion

$\alpha_P = \frac{\alpha_1+\alpha_2}{2}$

(10) **Answer :** (3)
Solution:
On increasing temperature drift velocity of electron in a metallic wire decrease
Conductivity of metallic wire decrease on increasing temperature

(11) **Answer :** (2)
Solution:

$P = \frac{V^2}{R} \qquad P \propto \frac{1}{R}$

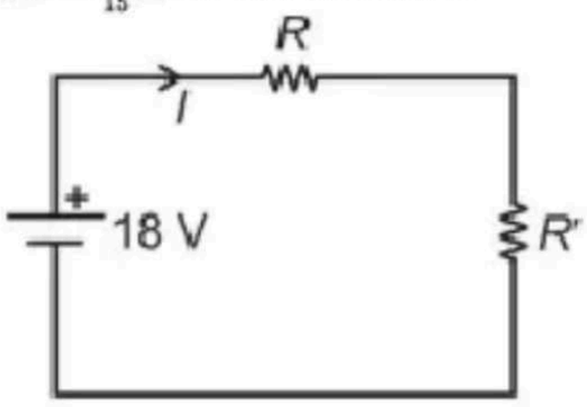
(12) **Answer :** (2)
Solution:
$i = neAv_d$.
Since current flowing through conductor is same. Hence Av_d = constant
$\because A_1 < A_2 < A_3$
$\therefore v_{d_1} > v_{d_2} > v_{d_3}$

(13) **Answer :** (2)
Solution:
From Ohm's law resistance $R = \frac{\Delta V}{\Delta I}$. Resistance is said to be negative if on increasing voltage or temperature, current decreases.

(14) **Answer :** (2)
Solution:
Last four resistors are in parallel.

$\frac{1}{R'} = \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{16} = \frac{4+4+4+3}{48}$

$R' = \frac{48}{15}\Omega$ Circuit becomes.



$I = \frac{18}{R+R'}$

$\frac{5}{2} = \frac{18}{R+\frac{48}{15}}$

$\therefore \quad R + \frac{48}{15} = \frac{18\times 2}{5}$

$R = \frac{36}{5} - \frac{48}{15} = \frac{108-48}{15} = \frac{60}{15} = 4\ \Omega$

(15) **Answer :** (2)
Solution:
Current density is a vector quantity having both direction and magnitude.

(16) **Answer :** (2)
Solution:

$i = \frac{20+20}{2+3+10} \Rightarrow i = \frac{8}{3}$ A

Potential difference across cell B

$V = E - ir$

$V = 20 - \frac{8}{3} \times 3 = 12$ V

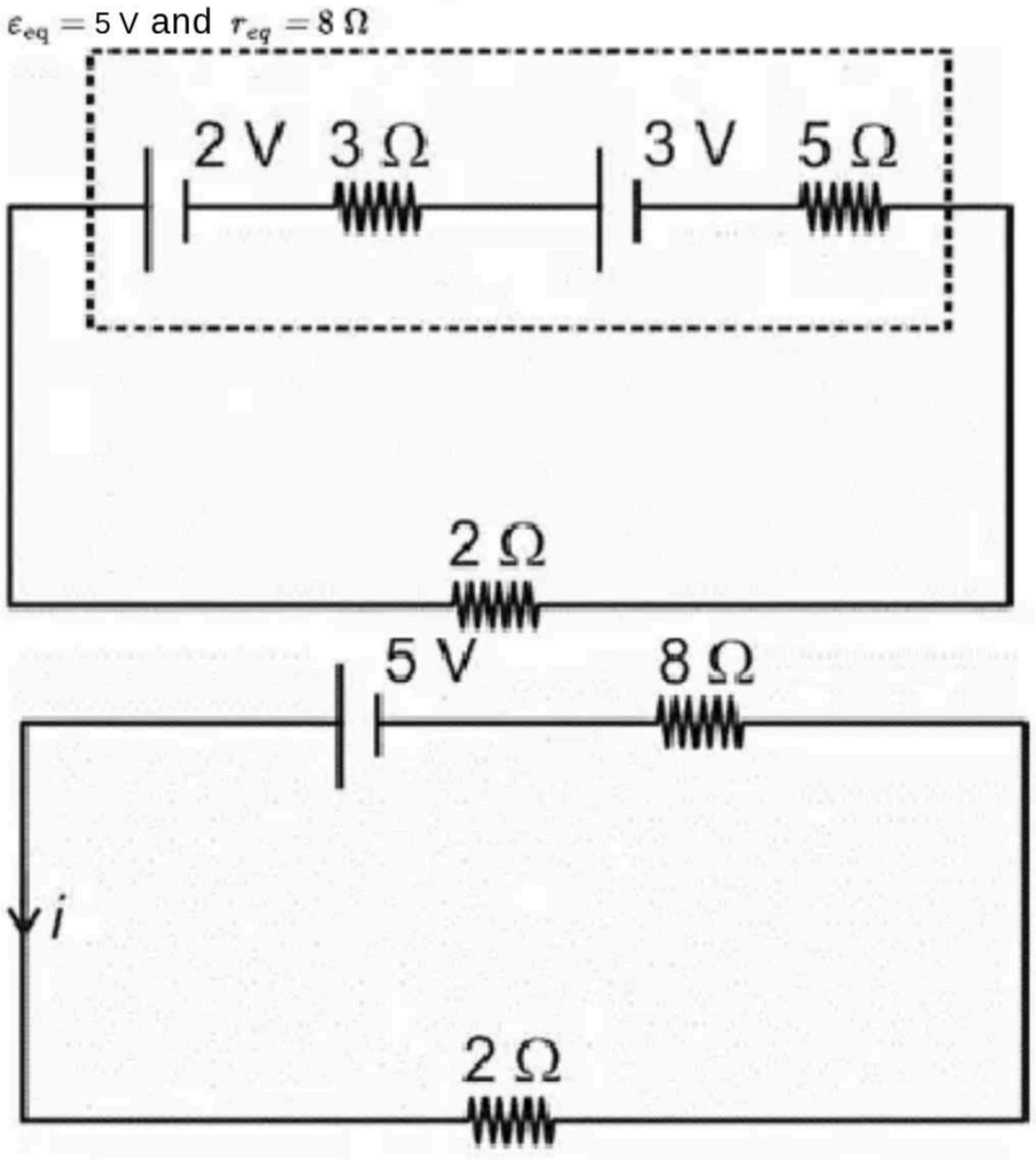
(17) Answer : (2)
Solution:
$J \propto E$
$J = \frac{I}{A}$
$H = I^2 \left(\rho \frac{I}{A}\right) t$
$I = neAv_d$

(18) Answer : (1)
Solution:
$\frac{P}{R} = \frac{Q}{S}, \frac{P}{Q} = \frac{R}{S}, PS = RQ$

(19) Answer : (3)
Solution:
As $\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$ and $r_{eq} = r_1 + r_2$

$\therefore \quad I = \frac{\varepsilon_1 + \varepsilon_2}{r_1 + r_2 + R}$

As the two cells are correctly connected in series,
$\varepsilon_{eq} = 5$ V and $r_{eq} = 8\ \Omega$

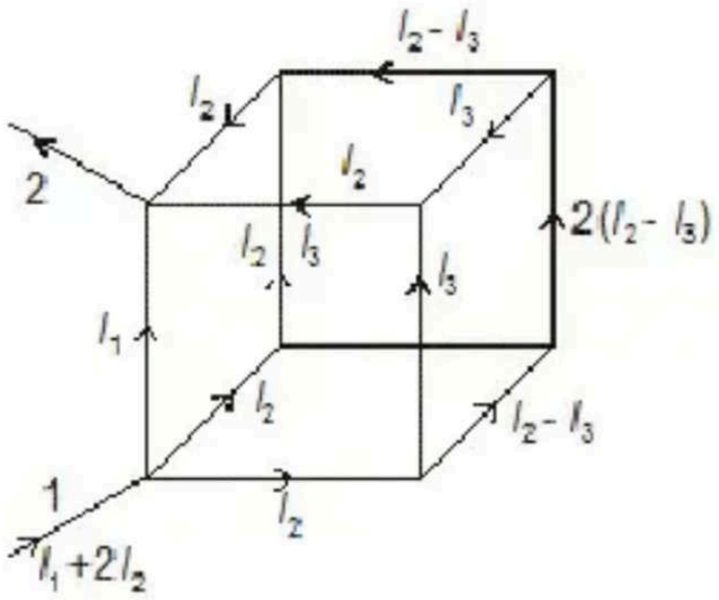


$\therefore$ Current in the loop, $i = \frac{5}{10} = 0.5$ A

(20) Answer : (2)
Solution:
$\because E_{eff} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = 3$ V

(21) Answer : (2)
Solution:
When S is closed, after long time steady state reaches and current through resistors
$i = \frac{20}{10}$ A $= 2$ A
$V_A - V_X = 2 \times 5 = 10$ V
$V_Y - V_C = 2 \times 5 = 10$ V

(22) Answer : (3)
Solution:
Between point 1 and 2

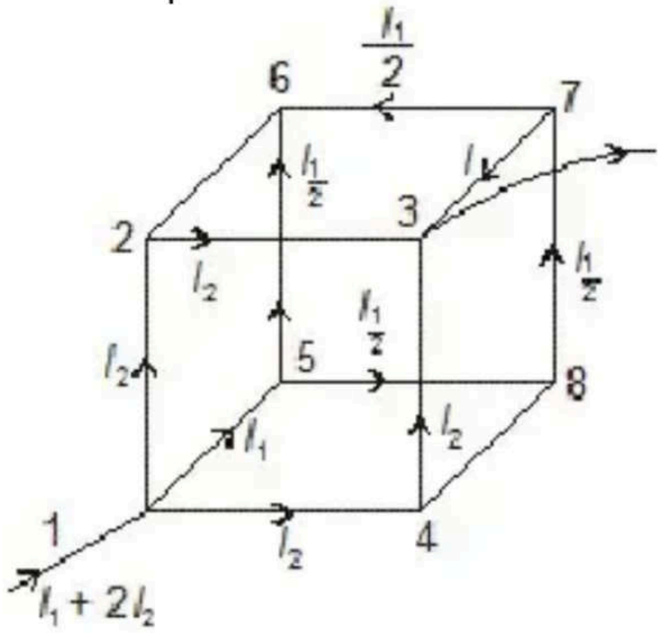


$(I_1 + 2I_2)\ R_{equ} = I_1 R$

$\frac{24}{5} I_2 R_{equ} = \frac{14}{5} I_2 R$

$R_{equ} = \frac{7}{12} R$

Between point 1 and 3

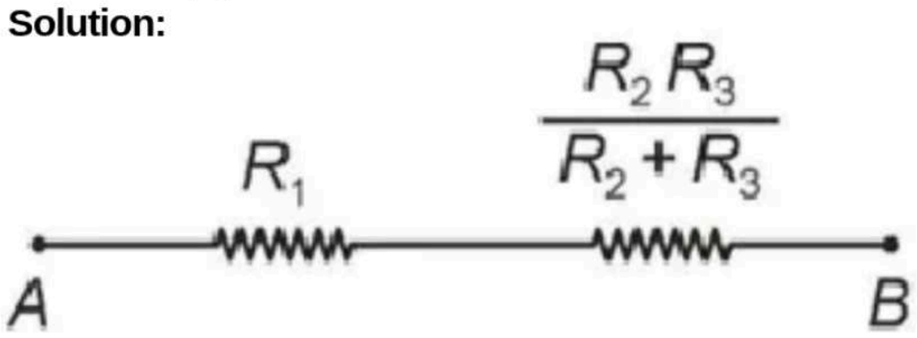


$(I_1 + 2I_2)\ R_{equ} = 2I_2 R$

$4I_1 R_{equ} = 3I_1 R \quad \left[\because I_2 = \frac{3I_1}{2}\right]$

$R_{equ} = \frac{3}{4} R$

(23) Answer : (3)
Solution:

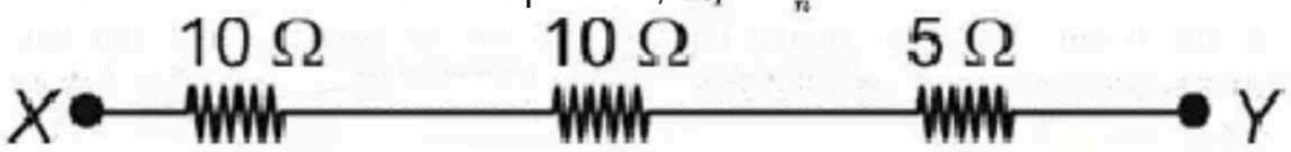


$R_{eq} = R_1 + \frac{R_2 R_3}{R_2 + R_3}$

(24) Answer : (2)
Solution:
Applying K.V.L in loop AEFC
$-5 + 5 - 12 \times i = 0$ or, $i = 0$

(25) Answer : (2)
Solution:
$V = E - ir$ …(i)
$i = \frac{E}{R+r}$ …(ii)
Solving we get
$r = \frac{(E-V)R}{V}$

(26) Answer : (3)
Solution:
For *n* identical resistances in parallel, $R_P = \frac{R}{n}$



$\therefore \quad R_{XY} = 25\ \Omega$

(27) Answer : (2)

Solution:

$P_{max} = \frac{E^2}{4r} = \frac{10\times10}{4\times2} = 12.5$ W

(28) Answer : (4)
Solution:
Both branches *AB* and *CD* are in parallel and have equal resistance.

(29) Answer : (2)
Solution:
Electric Current is a scalar quantity.

(30) Answer : (2)
Solution:

$i = \frac{10}{\frac{5}{2}+1} = \frac{20}{7}$ A

$v = E - ir = 10 - \frac{20}{7}\times1 = \frac{50}{7}$ V

(31) Answer : (1)
Solution:
Kirchhoff's voltage law is based on the conservation of energy.

(32) Answer : (3)
Solution:
$P = V_{rms}\, I_{rms} \cos\varphi$
$\cos\varphi = 1$

$P = \frac{\left(V_{rms}\right)^2}{R}$

$= \frac{40\times40}{8}$

= 200 W

(33) Answer : (3)
Solution:

Current through arm *AB*, $I_1 = \frac{10\text{ V}}{30\Omega}$

$I_1 = \frac{1}{3}$ A

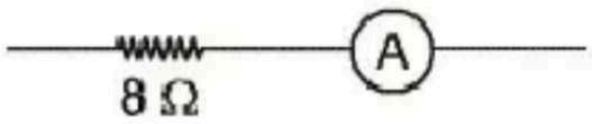
Current through arm *DC*, $I_2 = \frac{10\text{ V}}{30\Omega} = \frac{1}{3}$ A

Potential difference across *XY*:
$V_X - V_Y = (V_X - V_A) + (V_A - V_Y)$
$= -10I_1 + 20I_2$

$= -10\times\frac{1}{3} + 20\times\frac{1}{3}$

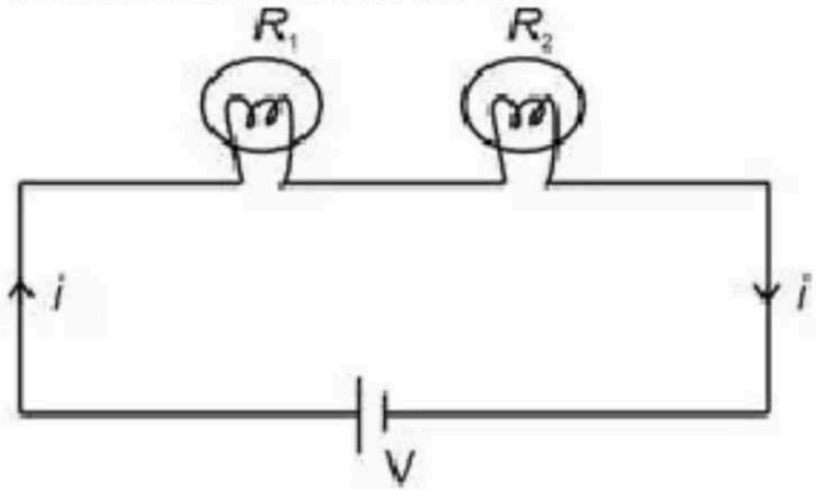
$= \frac{10}{3}$ V

(34) Answer : (1)
Hint:
Parallel circuit elements have same potential difference.
Solution:
Potential difference across resistance arm with ammeter is 32 V



Current is $\frac{32}{8} = 4$ A

(35) Answer : (3)
Solution:
In series same current will flow

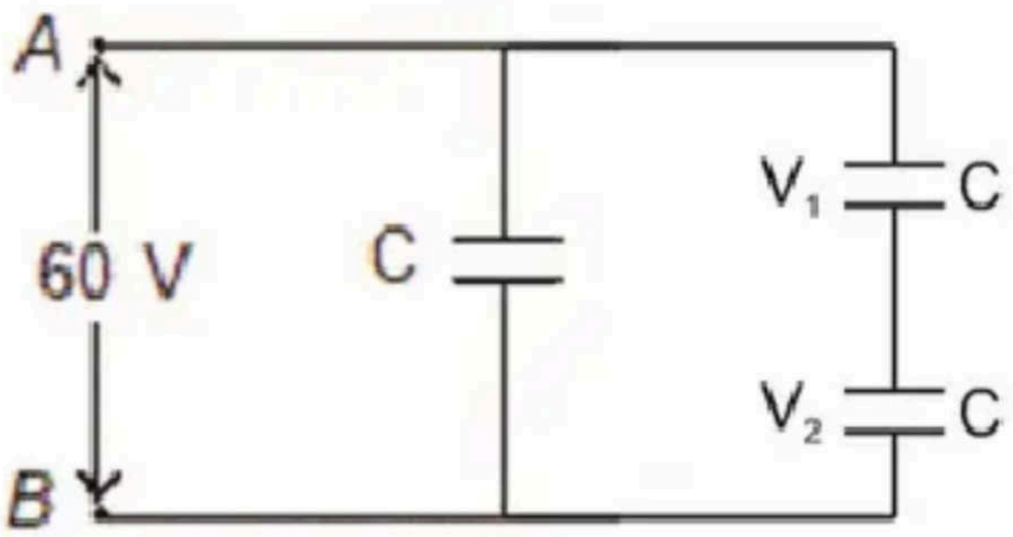


$P_1 = I^2R_1$

$P_2 = I^2R_2$

$\frac{P_1}{P_2} = \frac{R_1}{R_2} = \frac{2}{3}$

(36) Answer : (4)
Solution:
Both 2*C* are in series, therefore their effective will become *C*. Then



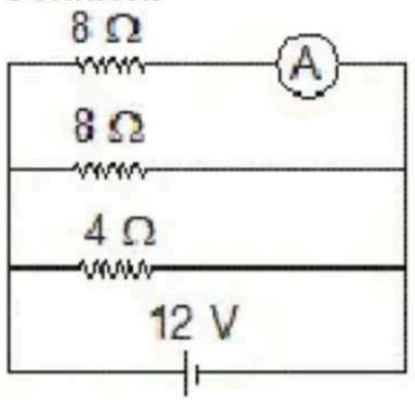
$V_1 = V_2 = 30$ V

(37) Answer : (3)
Solution:
$q = 3t^2 - t^3$
$i = 0$
$i = \frac{dq}{dt} = 6t - 3t^2$
$i = 0 \Rightarrow 3t(2 - t) = 0$
$t = 0$ or $t = 2$s
$q = 3(2)^2 - 2^3 = 4$ mC

(38) Answer : (4)
Hint:
$v_d = \frac{eE\tau}{m}$
Solution:
$J = nev_d = \frac{ne^2\tau}{m}E.$
$\frac{I}{A} = \frac{ne^2\tau}{m}\left(\frac{V}{L}\right)$
$V = \frac{m}{ne^2\tau}\frac{L}{A}I$
$\rho = \frac{m}{ne^2\tau}$

(39) Answer : (1)
Solution:
$t = \frac{dQ}{dt} = 6t + 2$
At $t = 2$, $i = 6 \times 2 + 2 = 14$ A

(40) Answer : (3)
Hint:
$I = \frac{V}{R}$
Solution:



Reading of ammeter will be current in that branch.
$I = \frac{12}{8} = \frac{3}{2}$ A

$\boxed{I = 1.5 \text{ A}}$

(41) Answer : (2)
Solution:
$\frac{\ell_1}{\ell_2} = \frac{100}{\frac{100 \times x}{100 + x}} = 3$
$\frac{100 + x}{x} = 3$

$x = 50\ \Omega$

(42) Answer : (4)
Solution:

$I = \frac{dq}{dt}$

$dq = \int_0^1 \left(2t^2 + t + 3\right) dt$

$= \left[\frac{2t^3}{3} + \frac{t^2}{2} + 3t\right]_0^1$

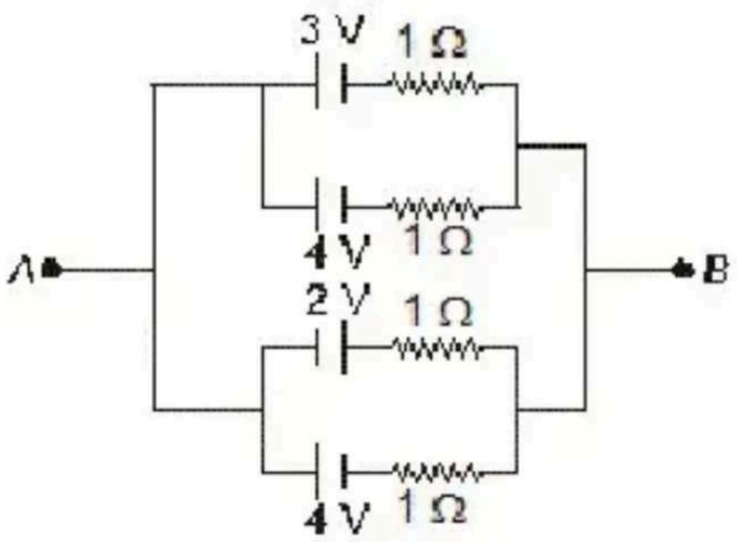
$= \left[\frac{2}{3} + \frac{1}{2} + \frac{3}{1}\right] = \left[\frac{4+3+18}{6}\right] = \frac{25}{6}\text{mC}$

(43) Answer : (2)
Solution:

Using: $E = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}$

For upper set: $\mathbf{E_1} = \frac{\frac{3}{1} + \frac{4}{1}}{\frac{1}{1} + \frac{1}{1}} = \frac{7}{2}$ V

For lower set: $\mathbf{E_2} = \frac{\frac{4}{1} - \frac{2}{1}}{\frac{1}{1} + \frac{1}{1}} = \frac{2}{2}$ V



$r_1' = \frac{1}{2}\Omega$

$r_2' = \frac{1}{2}\Omega$

$\mathbf{E}_{\text{equ}} = \frac{\frac{E_1}{r_1'} + \frac{E_2}{r_2'}}{\frac{1}{r_1'} + \frac{1}{r_2'}} = \frac{\frac{7/2}{1/2} + \frac{2/2}{1/2}}{2+2} = \frac{9}{4}\ \mathbf{V}$

(44) Answer : (4)
Solution:
The electron moves is conductor due to temperature even if there is no electric field, but in absence of external field net acceleration is zero.

(45) Answer : (2)
Solution:

$I = \frac{5}{8+2} = 0.5\ A$

Chemistry

(46) Answer : (4)
Hint:
Order of the reaction may not be equal to the number of molecule of reactant as seen from balanced equation.
Solution:
Order of the reaction may not be equal to the number of molecule of reactant as seen from balanced equation.

(47) Answer : (1)
Solution:

Rate = K[Conc.]^x
Let the hypothetical reaction be

$A(g) \rightarrow B(g)$

Now, $r_1 = K[A]^x$ ... (1)
where, x is order of reaction.
Now, when vol. of container is halved, conc. of A becomes double.

$r_2 = K[2A]^x$ ... (2)
Divide (2) by (1)

$\frac{r_2}{r_1} = \frac{K[2A]^x}{K[A]^x}$

(given $r_2 = 4r_1$)

$\therefore \quad 4 = [2]^x$

$\Rightarrow \quad \boxed{x = 2}$

(48) Answer : (2)

Solution:

Hydrolysis of ethyl acetate in acidic medium shows first order, but in basic medium follows second order.

(49) Answer : (1)

Solution:

	$A \rightarrow$	$2B +$	C
$t = 0$	100	0	0
$t = 300$	$100 - x$	$2x$	x

$P_{total} = 100 - x + 2x + x = 150$

$x = 25 \text{ mm} = \frac{25}{760} \text{ atm}$

t = 5 min = 300 s

$\text{rate} = \frac{x}{t} = \frac{25}{760 \times 300} = 1.1 \times 10^{-4} \text{ atm s}^{-1}$

(50) Answer : (2)

Hint:

Activation energy is the minimum amount of energy that is required to activate the molecules to a condition in which they can undergo chemical transformation.

Solution:

$K = A \cdot e^{-Ea/RT}$

$\ln K = \ln A - \frac{E_a}{RT}$

In case if $E_a = 0$

ln k = ln A

Where K = Rate constant

A = Frequency factor

(51) Answer : (2)

Solution:

$\ln k = \ln A - \frac{E_a}{R}\left(\frac{1}{T}\right)$

(52) Answer : (2)

Solution:

Rate = k (for zero order).

(53) Answer : (4)

Solution:

For a zero order reaction $t_{1/2} = \frac{[A_0]}{2k}$.

So, $t_{1/2} \propto [A_0]$

(54) Answer : (2)

Solution:

$2NH_3(g) \xrightarrow[Pt]{1130\,K} N_2(g) + 3H_2(g)$

$r = k[NH_3]^0$

It is a zero order reaction.

(55) Answer : (3)

Solution:

For a first order reaction $t_{x\%}$ remains constant.

Given for the first order reaction, $t_{40\%}$ = 50 sec.

Now $t_{64\%}$ is 2nd $t_{40\%}$ and for a first order reaction $t_{x\%}$ remains constant.

$\therefore \quad t_{64\%} = 2 \times t_{40\%} = 2 \times 50 = 100$ sec.

(56) Answer : (3)

Solution:

$\ln k = \underset{(\text{Slope})}{\left(\frac{-E_a}{R}\right)}\left(\frac{1}{T}\right) + \ln A$

(57) Answer : (2)

Solution:

Hint: For first order reaction, $k = \frac{0.693}{t_{1/2}}$

Sol.: $k = \frac{0.693}{30} = \frac{693 \times 10^{-3}}{30}$

$= \frac{69.3}{3} \times 10^{-3} = 23.1 \times 10^{-3}$

$= 0.023\ s^{-1}$.

(58) Answer : (3)

Solution:

$-\frac{1}{4}\frac{d[P]}{dt} = \frac{-d[Q]}{dt} = \frac{+1}{2}\frac{d[R]}{dt} = \frac{+1}{2}\frac{d[S]}{dt}$

Hence rate of appearance of R is twice the rate of disappearance of Q.

(59) Answer : (2)

Solution:

$H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$

$r_{av} = -\frac{\Delta[H_2]}{\Delta t} = -\frac{\Delta[Cl_2]}{\Delta t} = \frac{1}{2}\frac{\Delta[HCl]}{\Delta t}$

(60) Answer : (2)

Hint:

Answer (2)

Solution:

For first order reaction, $k = \frac{2.303}{t} \log \frac{a}{a-x}$

For 40% completion, $k = \frac{2.303}{50} \log \frac{100}{60}$

$\frac{0.693}{t_{1/2}} = \frac{2.303}{50} \log\left(\frac{5}{3}\right)$

$t_{1/2} = \frac{0.693 \times 50}{2.303 \times 0.23}$

$t_{1/2} = 65.4$ minutes $\simeq 65$ minutes

(61) Answer : (1)

Hint:

Unit of rate = mol L^{-1} s^{-1}

Solution:

r = k [A]

(mol L^{-1} s^{-1}) = k (mol L^{-1})

k = s^{-1}

(62) Answer : (3)

Solution:

$\log \frac{k_{cat}}{k_{uncat}} = \frac{1}{2.303RT}\left[E_{a(uncat)} - E_{a(cat)}\right]$

$\log \frac{k_{cat}}{k_{uncat}} = \frac{1}{2.303 \times 2 \times 310}\left[(200 \times 10^3) - (60 \times 10^3)\right]$

$\log \frac{k_{cat}}{k_{uncat}} = 98$

(63) Answer : (2)

Solution:

Decomposition of N_2O_5 is a first order reaction.

(64) Answer : (2)

Solution:

Rate law = k[H_2O_2]

(65) Answer : (1)

Solution:

Answer (1)

	$2N_2O_5 \longrightarrow$	$4NO_2 +$	O_2
$t = 0$	p°	0	0
time t,	$p^\circ - x$	$2x$	$\frac{x}{2}$
∞time,	$p^\circ - p^\circ$	$2p^\circ$	$\frac{p^\circ}{2}$

at ∞ time, $p_{total} = 2p^{\circ} + \frac{p^{\circ}}{2} = 500 \therefore p^{o} = 200$

at 40 min, $p_{total} = p^{0} + \frac{3x}{2} = 350 \quad \therefore x = 100$

$\therefore k = \frac{2.303}{40} \log \frac{200}{200-100} = 0.017 \text{ min}^{-1}$

(66) Answer : (1)

Solution:

$t = \frac{2.303}{k} \log \frac{1}{0.000001}$

$t = \frac{2.303}{k} \log 10^{6} \Rightarrow \frac{6 \times 2.303}{k}$

(67) Answer : (1)

Hint:

Unit of k = $(\text{mol/lit})^{1-n}\ s^{-1}$

Solution:

For zero order reaction, unit of k = mol $L^{-1}\ s^{-1}$ Unit of rate = mol $L^{-1}\ s^{-1}$

(68) Answer : (1)

Solution:

Modified Arrhenius equation is

$k = PZ_{AB}e^{-E_a IRT}$

(69) Answer : (4)

Hint:

Stronger is the base, higher will be the equilibrium constant for protonation.

Solution:

Among the given compounds, only pyridine, has localized lone pair of electron.

∴ It is most basic. As a result it has the maximum value of equilibrium constant for protonation.

(70) Answer : (4)

Solution:

Rate of disappearance of B

$= -\frac{d[B]}{dt} = -\frac{1}{6}\frac{d[A]}{dt} = \frac{1}{9}\frac{d[C]}{dt} = \frac{2}{3}\frac{d[D]}{dt}$

(71) Answer : (1)

Solution:

$A_t = A_0 - kt$ {For zero order reaction}

(72) Answer : (2)

Solution:

188 g AgBr contains 80 g bromine

1.88 g of AgBr contains = $\frac{80}{188} \times 1.88 = 0.8\ g$

Percentage of bromine = $\frac{0.8}{1.6} \times 100 = 50\%$

(73) Answer : (4)

Solution:

$A(g) \rightarrow B(g) + C(g)$

	$A_{(g)}$	$\longrightarrow$	$B_{(g)}$	+	$C_{(g)}$
$t = 0$	10		–		–
$t = 20$ min	$10 - P$		P		P

P_T = 12 atm

12 = (10 – P) + P + P

12 = 10 + P

$\boxed{P = 2}$

$kt = 2.303 \log\left(\frac{10}{10-2}\right)$

$\therefore k = \frac{2.303}{20} \log\left(\frac{10}{8}\right)$

(74) Answer : (3)

Solution:

$r = k[A]^{x}[B]^{y}$

$3.2 \times 10^{-4} = k\,[0.50]^{x}\,[0.50]^{y}$ ...(i)

$3.2 \times 10^{-4} = k\,[1.00]^{x}\,[0.50]^{y}$ ...(ii)

$1.28 \times 10^{-3} = k\,[1.00]^{x}\,[1.00]^{y}$ ...(iii)

From equation (i), (ii) and (iii)

x = 0, y = 2

$r = k\,[B]^{2}$

(75) Answer : (2)
Solution:

$$\frac{\left(t_{1/2}\right)_1}{\left(t_{1/2}\right)_2} = \left(\frac{a_2}{a_1}\right)^{n-1}$$

$$\frac{75}{150} = \left(\frac{2}{1}\right)^{n-1}$$

$\Rightarrow 2^{-1} = 2^{n-1}$
$-1 = n - 1$
$\Rightarrow n = 0$
So it is zero order reaction.

(76) Answer : (4)

(77) Answer : (2)
Hint:
Rate of reaction = $k[A]^x[B]^y[C]^z$
Solution:
$5 \times 10^{-5} = k(0.01)^x(0.005)^y(0.01)^z$ ....(1)
$5 \times 10^{-5} = k(0.02)^x(0.005)^y(0.01)^z$ ....(2)
$2.5 \times 10^{-5} = k(0.01)^x(0.01)^y(0.01)^z$ ....(3)
$20 \times 10^{-5} = k(0.01)^x(0.005)^y(0.02)^z$ ....(4)
By solving (1) and (2), x = 0
By solving (1) and (3), y = –1
By solving (1) and (4), z = 2
$\therefore$ Rate of reaction $= k[A]^0[B]^{-1}[C]^2$

(78) Answer : (1)
Solution:
Arrhenius proposed the following empirical equation relating the rate constant k and temperature T (in Kelvin),
$k = A.\ e^{-E_a/RT}$

(79) Answer : (3)
Solution:
Hydrolysis of ester with NaOH is second order reaction.

(80) Answer : (2)

(81) Answer : (3)
Solution:
Unit of rate constant.
For n^{th} order = $(mol)^{1-n}\ Ln^{-1}\ s^{-1}$

(82) Answer : (2)

(83) Answer : (4)
Solution:
$t = nt_{1/2}$
$800 = n \times 160 \Rightarrow n = 5$

$$a_t = \frac{a_0}{2^n} = \frac{a_0}{2^5} = \frac{1}{32}$$

(84) Answer : (4)
Solution:
Specific reaction rate is rate constant of a reaction which depends on temperature of reaction.

(85) Answer : (3)
Hint:
If a compound is in excess, then order w.r.t. that compound will be zero
Solution:
Overall order = 1 + 1 = 2

(86) Answer : (3)
Solution:

$$\frac{r_2}{r_1} = \frac{P_{A_2}.P_{B_2}^2}{P_{A_1}.P_{B_1}^2} = \frac{0.1\times(0.4)^2}{0.4\times1^2} = \frac{1}{25}$$

(87) Answer : (2)
Solution:
For first order reaction, $k = \frac{2.303}{t} \log \frac{a}{a-x}$

For 40% completion, $k = \frac{2.303}{50} \log \frac{100}{60}$

$\frac{0.693}{t_{1/2}} = \frac{2.303}{50} \log \left(\frac{5}{3}\right)$

$t_{1/2} = \frac{0.693 \times 50}{2.303 \times 0.23}$

$t_{1/2}$ = 65.4 minutes $\simeq$65 minutes

(88) Answer : (1)

(89) Answer : (2)
Hint:

$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$

Solution:

$\log\left(\frac{2r}{r}\right) = \frac{E_a}{2.303 \times 8.314} \times \left(\frac{10}{290 \times 300}\right)$

$E_a = 50.140 \times 10^3 \simeq 50 \text{ kJ mol}^{-1}$

(90) Answer : (4)
Solution:
Hint: Addition of a catalyst provides new path to reaction.
Sol.: Enthalpy of the reaction $\Delta_r H = (E_a)_f - (E_a)_b$. Catalyst does not alter $\Delta_r H$, as it lowers both $(E_a)_f$ and $(E_a)_b$ by equal amounts.

Botany

(91) Answer : (2)
Solution:
Such males have gynaecomastia.

(92) Answer : (3)
Solution:
Dominant gene expresses in both homozygous and heterozygous conditions.
Recessive gene express only in pure or homozygous condition.

(93) Answer : (3)
Solution:
A dominant allele expresses in both homozygous and heterozygous conditions.
A recessive allele expresses only in homozygous condition.

(94) Answer : (1)
Solution:
Dominant epistasis is shown by fruit colour in summer squash.

(95) Answer : (3)
Solution:
Grasshopper males have one chromosome less than females.

(96) Answer : (1)
Solution:
Mendel worked with 14 true-breeding pea plant varieties (7 pairs of varieties) and 7 characters during his experiments.

(97) Answer : (3)
Solution:
Mendel selected 14-true breeding pea plant varieties for his experiments.

(98) Answer : (3)
Solution:
(i) Dominant gene expresses in both homozygous and heterozygous conditions.
(ii) Recessive gene express only in pure or homozygous condition.

(99) Answer : (3)
Solution:
In a family, the father has the B blood group and mother has the A blood group. If their children show 50% probability of AB blood group, it indicates thateither of the parents is heterozygous.

(100) Answer : (1)
Solution:
Mendel selected Pea plant for experiments because, Pea plant has many distinct alternative traits, produces a large number of seeds and they are true breeding plants.

(101) Answer : (2)

Solution:
Bell-shaped curve.

(102) Answer : (3)
Solution:
Law of independent assortment can be studied by dihybrid cross.

(103) Answer : (2)
Solution:
Gregor Johann Mendel was the first to apply statistical methods and mathematical logics for analysis of experimental results.

(104) Answer : (1)
Solution:
Sterile females with rudimentary ovaries and 44 + X0 genetic compliment suffer from Turner's syndrome.

(105) Answer : (1)
Solution:
When a single gene can exhibit multiple phenotypic expressions, is called a pleiotropic gene. This is due to the effect of a gene on metabolic pathways which contribute towards different phenotypes.

(106) Answer : (4)
Solution:
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(107) Answer : (1)
Solution:
To calculate no. of gametes = 2^n
Where n $\Rightarrow$ degree of heterozygosity
n = 3
$\Rightarrow 2^n = 2^3 = 8$

(108) Answer : (4)
Solution:
The given pedigree represents that the trait is inherited as autosomal recessive disorder *i.e.* cystic fibrosis.

(109) Answer : (1)
Solution:
Multiple alleles can be studied in population.

(110) Answer : (3)
Solution:
Nine.

(111) Answer : (3)
Solution:
Dominant phenotype can appear in homozygous and heterozygous condition.

(112) Answer : (4)
Solution:
A gene can have more than 2 alleles.

(113) Answer : (3)
Solution:
Autosomal recessive.

(114) Answer : (1)
Solution:
Translocation between chromosomes 22 & 9.

(115) Answer : (1)
Solution:
Gregor Mendel conducted hybridisation experiment on garden pea for 7 years. This period was 1856 – 1863

(116) Answer : (3)
Solution:
50% daughters can be carriers.

(117) Answer : (2)
Hint:
Genotype of child must be = aa
Solution:
Genotype of parents should be = Aa, Aa
$\frac{1}{4}$

(118) Answer : (2)
Solution:

In a monohybrid cross, law of dominance is used to explain the expression of only one of the parental traits in the F_1 generation and the expression of both in the F_2. It also explains the proportion of 3:1 obtained in F_2 generation.

(119) Answer : (1)

Hint:

Types of gametes formed are 2^n, where "n" is number of hybrid.

Solution:

In PpQqRr × ppqqrr, probability of getting $\mathrm{Pp} = \frac{1}{2}, \mathrm{qq} = \frac{1}{2} \text{and } \mathrm{Rr} = \frac{1}{2}$

So, probability of getting genotype PpqqRr is $= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$

(120) Answer : (4)

Solution:

Mendel selected 14 true breeding pea varieties. His experiment had a large sampling size, which gave great credibility to the data that he collected.

(121) Answer : (4)

Solution:

The pedigree represents autosomal recessive disorder that is cystic fibrosis.

(122) Answer : (1)

Solution:

In homozygous condition dominant & recessive traits both are expressed. But in heterozygous only dominant trait is expressed.

(123) Answer : (3)

Solution:

Heterozygous individuals appear apparently unaffected.

(124) Answer : (2)

Solution:

Due to non-disjunction of chromosomes.

(125) Answer : (1)

Solution:

In F_2 generation of Mendelian dihybrid cross (YYRR X yyrr) yellow cotyledon and round seeded plants were obtained in 1 : 1 frequency.

(126) Answer : (2)

Solution: Answer (2)

(127) Answer : (1)

Solution:

Sterile females with rudimentary ovaries and 44 + X0 genetic compliment suffer from Turner's syndrome.

(128) Answer : (2)

Solution:

Guanine is converted to xanthine which also pairs with cytosine.

(129) Answer : (3)

Solution:

In a family, the father has the B blood group and mother has the A blood group. If their children show 50% probability of AB blood group, it indicates that either of the parents is heterozygous.

$I^BI \times I^AI^A$

$I^AI^B, I^AI, I^AI^A, I^AI^B,$

AB, A, A, AB.

(130) Answer : (3)

Solution:

Terminal flower position in pea plant is a recessive trait, so it is expressed only in homozygous condition.

(131) Answer : (2)

Solution:

Mother should at least be a carrier and father haemophilic.

(132) Answer : (3)

Hint:

Number of phenotypes depends on number of gametes produced.

Solution:

Since 1 parent in all options is aabb

(test cross). Thus types of gamete will depend only on 2nd parent.

AABb × aabb, AABB × aabb, AaBb × aabb

↓ ↓ ↓

2 types of gametes | 1 type of gametes | 4 types of gametes

$$\underbrace{AaBB}_{\downarrow} \times aabb$$

2 types of gametes

(133) Answer : (2)
Solution:
Phenylketonuria is an autosomal recessive condition.

(134) Answer : (3)
Solution:
Terminal flower position in pea plant is a recessive trait, so it is expressed only in homozygous condition.

(135) Answer : (1)
Solution:
To all daughters.

Zoology

(136) Answer : (2)
Solution:
Relaxin is secreted from ovary at the end of gestation period.

(137) Answer : (3)
Solution:
Census is conducted by a separate agency or body under Government of India.

(138) Answer : (2)
Solution:
Lippes loop is a non-medicated IUD.

(139) Answer : (4)
Hint:
Withdrawal of penis is a natural contraceptive method.
Solution:
Coitus interruptus (withdrawal method) is a natural method of contraception, in which withdrawal of penis is ensured before ejaculation. Surgical method or sterilisation procedure involves vasectomy or tubectomy.

(140) Answer : (3)
Solution:
Progesterone alone or in combination with oestrogens are given in oral pills as contraceptives.

(141) Answer : (2)
Solution:
Diaphragms, cervical caps and vaults are barriers made of rubber, inserted in the female reproductive tract to cover the cervix during coitus.

(142) Answer : (2)
Solution:
IUDs are the most widely accepted methods of contraception in India. CuT is an IUD (Intra Uterine Device).

(143) Answer : (3)
Hint:
Mulberry like.
Solution:

$$\underset{\text{(one cell)}}{\text{Zygote}} \longrightarrow \underset{(8-16\text{cells})}{\text{Morula}} \longrightarrow \text{Blastocyst}$$

(144) Answer : (1)
Solution:
Saheli contains non-steroidal preparations and it is a ‘once a week’ a pill.

(145) Answer : (3)
Solution:
Human placental lactogen, *i.e.*, hPL acts on mammary glands to facilitate the process of lactation. Cortisol, progesterone and estrogen are secreted in non-pregnant females also.

(146) Answer : (2)
Solution:
Government of India legalized MTP in 1971 with some strict conditions to avoid its misuse.

(147) Answer : (4)
Solution:

Saheli–the new oral contraceptive for the females contains a non-steroidal preparation. It is a 'once a week' pill with very few side effects and high contraceptive value.

(148) Answer : (4)
Solution:
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(149) Answer : (1)
Solution:
Condoms, diaphragms, cervical caps and vaults are barriers for birth control. Use of condoms has increased in recent years due to its additional benefits of protecting the users from contracting STI's.

(150) Answer : (3)
Solution:
The world population was around 6 billion by 2000.

(151) Answer : (3)
Solution:
Emergency contraceptives are administered within 72 hours of unprotected sex or in rape case.

(152) Answer : (1)
Solution:
The hormone releasing IUDs make the uterus unsuitable for implantation and cervix hostile to sperms.

(153) Answer : (2)
Solution:
An ideal contraceptive is user friendly, inexpensive, effective, reversible and does not interfere with the sexual drive (libido) of a person.

(154) Answer : (3)
Solution:
Amniocentesis is based on analysis of amniotic fluid which contains cells of foetus.

(155) Answer : (3)
Hint:
Identify inert and hormone releasing.
Solution:
Copper releasing IUDs – CuT, Cu7, Multiload 375
Hormone releasing IUDs – Progestasert, LNG-20
Non-medicated IUDs – Lippes loop

(156) Answer : (4)
Hint:
Progesterone gives negative feedback to hypothalamus.
Solution:
Higher concentration of estrogen and progesterone during luteal phase give negative back to hypothalamus, thereby inhibiting the release of GnRH from hypothalamus and LH, FSH from pituitary.

(157) Answer : (3)
Solution:
Oral pills inhibit ovulation and implantation.

(158) Answer : (4)
Hint:
Surgical method is the permanent method of family planning.
Solution:
Oral steroidal contraceptive pills alter the quality of cervical mucus. They also inhibit ovulation and implantation.
Surgical methods affect gamete transport.

(159) Answer : (2)
Solution:
The combined pills are taken for 21 days, starting preferably within the first five days of menstrual cycle.

(160) Answer : (1)
Solution:
Acrosome contains lytic enzymes which help the sperm to penetrate ovum.

(161) Answer : (4)
Solution:
'Saheli' is once is a week pill i.e. taken once in a week. It is non steroidal.

(162) Answer : (3)
Solution:
Lactational amenorrhoea is caused or induced by prolactin which suppresses GnRH secretion so gonadotrophins are suppressed.

(163) Answer : (1)
Solution:

Condoms and vasectomy are contraceptive method in male.

(164) Answer : (2)

Solution:

Spermicide with cervical cap act synergistically to increase contraceptive effectiveness

(165) Answer : (3)

Solution:

Barrier methods block the entry of sperms through the cervix.

(166) Answer : (1)

(167) Answer : (2)

Solution:

Gossypol inhibits spermatogenesis.

(168) Answer : (2)

Solution:

Fast blockage – Depolarisation

Slow blockage – Cortical and zona reactions

(169) Answer : (3)

Hint:

Bleeding during reproductive cycle is called menses.

Solution:

The menstrual cycle consists of four phases

(1) Menstrual (Bleeding) phase

(2) Follicular (Proliferative) phase

(3) Ovulatory phase

(4) Luteal (Secretory) phase

(170) Answer : (2)

Hint:

Consumption of oral contraceptive pills starts during Ist phase of menstrual cycle.

Solution:

Oral contraceptive pills are consumed during first five days of menstrual cycle but nonsteroidal oral contraceptive pills like Saheli is taken only once in a week.

(171) Answer : (2)

Hint:

It is called fertile period.

Solution:

During periodic abstinence, couples abstain from coitus from day 10-17 of menstrual cycle, when ovulation is expected.

(172) Answer : (3)

Solution:

Demographic transition is the period when birth rate equals the death rate.

(173) Answer : (2)

Solution:

IUDs are intrauterine devices fitted into the uterus.

(174) Answer : (1)

Solution:

Multiload 375 is a copper releasing IUD or intra uterine device that has 375 mm^2 of copper wire wound around its stem.

(175) Answer : (3)

Solution:

Rheumatic heart disease *i.e.* damage to heart valves occurs following an infection of throat by *Streptococcus pyogenes*.

(176) Answer : (3)

Solution:

Chemical method involves use of spermicidal chemicals while the barrier method involves use of devices that prevent the gametes from meeting.

(177) Answer : (3)

Solution:

Mifepristone acts as a competitive progesterone receptor antagonist. Its binding affinity for glucocorticoid receptor is more than ten times that of cortisol.

(178) Answer : (3)

Solution:

Identical twins are monozygotic. Each blastomere does not give rise to an embryo.

(179) Answer : (3)

Solution:

Saheli is non steroidal contraceptive which contains centrchoman affecting implantation.

(180) Answer : (2)

Solution:

Progesterone is thermogenic in nature.